

Title of research:

How does creating an Adaptive Learning Zone Using Artificial Intelligence (ALZUAI) serve as a holistic monitoring and response framework for individuals with Alzheimer's Disease?

Abstract

Alzheimer's Disease (AD), currently the most common cause of dementia, is characterized by the accumulation of amyloid plaques and neurofibrillary tangles in the brain, leading to neuronal degeneration and brain atrophy. Patients with AD experience progressive cognitive and behavioral impairments, including memory loss, decreased concentration, impaired reasoning, and difficulties with daily functioning. Globally, over 55 million people are living with dementia, with AD accounting for 60-70% of cases. This places a substantial burden on caregivers, who often experience emotional stress, depression, anxiety, and a reduced quality of life. To address this gap, this study explores the development of an assistive ecosystem for Alzheimer's care: Adaptive Learning Zone Using Artificial Intelligence (ALZUAI). ALZUAI integrates voice assistants, motion sensors, environmental cues, and fall detection, providing real-time task reminders, safety alerts, and gentle guidance, all without requiring wearable devices within the home, a known barrier in AD care. Many existing smart-home dementia studies focus on issue detection rather than active assistance, something ALZUAI would provide, i.e, prompts, guidance, notifications. Machine learning personalizes reminders and notifications based on behavioral trends, and caregivers receive updates via SMS or mobile application. By offering continuous, unobtrusive monitoring and proactive support, ALZUAI aims to reduce caregiver oversight burden, enhance patient safety and improve overall quality of life for both patient and caregivers.

Similar research

Jiang, D (2025). Artificial intelligence-driven personalized space design and implementation in the aging-friendly renovation of smart home. Discover Artificial Intelligence, 5.
<https://link.springer.com/article/10.1007/s44163-025-00556-7>

Shaik, M. Anik, F. Hasan, M. Chakravarty, S. Ramos, M. Rahman, M. Ahamed, S. Sakib, N.(2025). Advancing Remote Monitoring for Patients With Alzheimer Disease and Related Dementias: Systematic Review. JMIR Aging, 8. <https://aging.jmir.org/2025/1/e69175/>

Qiao, Y. Wang, L. Liu, Y. Chen, W. (2025). The adoption of assistive smart home technology and health-related quality of life among older adults: the moderating role of depressive symptoms. Frontier Public Health, 13.
<https://www.frontiersin.org/articles/10.3389/fpubh.2025.1726983/full>

Giansanti, D. (2025). Integrating AI and Assistive Technologies in Healthcare: Insights from a Narrative Review of Reviews. MDPI. Healthcare, 13 (5), 556. <https://www.mdpi.com/2227-9032/13/5/556>